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Influence of spanwise flexibility on steady and dynamic responses of airfoils vs hydrofoils

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ABSTRACT

There is growing interest in using lighter and more flexible foils with active and/or passive smart actuation mechanisms to increase efficiency and maneuverability. For efficient operations, airfoils and hydrofoils tend to have an effective aspect ratio greater than two, which permits spanwise flexibility depending on the choice of material and architectural design. The fluid density plays an important role in the steady and dynamic performance, including the governing hydroelastic instability mechanism. To understand the influence of spanwise flexibility, we examine the response of a rectangular cantilevered foil as a canonical proxy for more complex lift generating devices. This research aims to investigate the influence of key geometric, material, and flow parameters on the steady and dynamic responses of the spanwise flexible airfoils vs hydrofoils. The results are obtained using an inviscid frequency- and time-domain fluid-structure interaction model that accounts for flow-induced bend-twist coupling terms and are compared with inviscid theory results without the flow-induced bend-twist coupling terms. The results show that the flow-induced bend-twist coupling effects impact the natural frequencies and damping coefficients, and such effects grow with increasing fluid density and flow speed. Hence, the flow-induced bend-twist coupling effects are more critical for hydrofoils than airfoils, particularly for the twisting mode. Ignoring the flow-induced bend-twist coupling terms leads to an incorrect prediction of flutter speed and over-prediction of the damping, which is dangerous because the actual vibrations and dynamic load amplification may be higher than the prediction, which could lead to accelerated fatigue. The results demonstrate that flexible hydrofoils have much lower natural frequencies and higher damping coefficients than airfoils due to higher fluid inertial and damping forces, both of which are proportional to the fluid density. While all components of the fluid forces are proportional to the fluid density, the fluid damping forces grow with the velocity, and the fluid disturbing forces grow with velocity square. Therefore, divergence tends to be the governing instability mode for hydrofoils, while flutter is typically the governing instability mode for airfoils. The maximum aero/hydro-elastic bending and twisting deformations are limited by the corresponding reduced stable velocity to avoid divergence or flutter.

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NOMENCLATURE

$\mathbf{A}$	Constant vector that depends on the initial bending and twisting deformations
A	Non-dimensional distance from mid-chord to elastic axis, positive for elastic axis after mid-chord
AR	Aspect ratio for cantilevered rectangular configurations, AR $= 2s/c$
B	Semi-chord length, $b = c/2$ (m)
C_L	Lift coefficient, positive for upwards
C_{L_α}	The slope of the lift coefficient, positive for upwards, $C_{L_\alpha} = dC_L/d\alpha_{\text{eff}}$
C_M	Moment coefficient, positive for counter-clockwise about the elastic axis (EA)
$C_{M,AC}$	Moment coefficient about the aerodynamic center, positive for counter-clockwise about the elastic axis (EA)
$C(k)$	Theodorsen's circulation function
$\tilde{\mathbf{C}}_f$	Non-dimensional generalized fluid damping matrix

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$\tilde{\mathbf{C}}_s$ Non-dimensional generalized structural damping matrix
$\tilde{C}_{s,h}$ Generalized structural bending damping value, $2\tilde{m}\omega_h\zeta_{s,h}$ (kg/s)
$\tilde{C}_{s,\theta}$ Generalized structural twisting damping value, $2\tilde{I}_\theta\omega_\theta\zeta_{s,\theta}$ (kg m^2/s)
c Chord length (m)
D Non-dimensional distance from elastic axis to three-quarter-chord ($3c/4$) ($d = a - 0.5$)
E Non-dimensional distance from elastic axis to aerodynamic center (AC), positive for AC upstream of the elastic axis ($e = a + 0.5$)
E_s Young's modulus (Pa)
$\tilde{\mathbf{F}}_{\text{dynamic}}$ Non-dimensional generalized dynamic fluid forces
$\tilde{\mathbf{F}}_{\text{fluid}}$ Non-dimensional generalized fluid forces
$\tilde{\mathbf{F}}_{\text{steady}}$ Non-dimensional generalized steady fluid forces
$f(z)$ Spanwise bending shape function (m)
f_h^* Natural bending frequency, $f_h^* = \omega_h^*/(2\pi)$ (Hz)
f_θ^* Natural twisting frequency, $f_\theta^* = \omega_\theta^*/(2\pi)$ (Hz)
$\bar{f}$ Non-dimensional frequency, $\bar{f} = f/f_\theta$
$\overline{f_h}$ Non-dimensional natural bending frequency, $\overline{f_h} = f_h^*/f_\theta$
$\overline{f_\theta}$ Non-dimensional natural twisting frequency, $\overline{f_\theta} = f_\theta^*/f_\theta$
$G(z)$ Spanwise twisting shape function (°)
H Instantaneous bending deformation, positive for upwards (m)
$H_1^2(k)$, $H_0^2(k)$ H ä nkel functions, i.e., Bessel functions of the third kind
$\tilde{h}$ Spanwise bending deformation, positive for upwards (m)
$\tilde{h}_{\text{elastic}}$ Aero/hydroelastic bending deformation (m)
$\tilde{I}_\theta$ Generalized structural mass moment of inertia, $\tilde{m}r_\theta^2 b^2$ (kg m^2)
$\tilde{K}_{s,h}$ Generalized structural bending stiffness value, $\tilde{m}\omega_h^2$ (N/m)
$\tilde{K}_{s,\theta}$ Generalized structural twisting stiffness value, $\tilde{I}_\theta\omega_\theta^2$ (N m)
$\tilde{\mathbf{K}}_f$ Non-dimensional generalized fluid disturbing matrix
$\tilde{\mathbf{K}}_s$ Non-dimensional generalized structural restoring matrix
k Reduced frequency: the ratio between structural oscillation frequency and fluid convection frequency, $k = \omega b/U$
$\tilde{L}_{\text{dynamic}}$ Non-dimensional generalized dynamic fluid lift force, positive for upwards
$\tilde{L}_{\text{steady}}$ Non-dimensional generalized steady fluid lift force, positive for upwards
$\tilde{\mathbf{M}}_f$ Non-dimensional generalized fluid inertial matrix
$\tilde{\mathbf{M}}_s$ Non-dimensional generalized structural inertial matrix
$\tilde{M}_{\text{dynamic}}$ Non-dimensional generalized dynamic fluid moment force, positive for counter-clockwise about the elastic axis (EA)
$\tilde{M}_{\text{steady}}$ Non-dimensional generalized steady fluid moment force, positive for counter-clockwise about the elastic axis (EA)
$\tilde{m}$ Generalized structural mass (kg)
Re Reynolds number: the ratio between fluid inertial force and fluid viscous force, $\text{Re} = Uc/\nu_f$
r_θ Non-dimensional radius of gyration about the elastic axis, $r_\theta = \sqrt{\tilde{I}_\theta/(\tilde{m}b^2)}$
S_f, S_g, S_{ff}, S_{fg}, S_{gg} Integrated shape functions
s Span length (m)
t Time (s)
$\bar{t}$ Non-dimensional time, $\bar{t} = t\omega_\theta$
U Inflow velocity (m/s)
U_d Static divergence velocity (m/s)
$\bar{U}$ Reduced velocity: the ratio between fluid convection frequency and structural first in-air natural twisting frequency, $\bar{U} = U/(\omega_\theta b)$
$\bar{U}_{\text{stable}}$ Reduced stable velocity, $\bar{U}_{\text{stable}} = U/U_d$
$\tilde{\mathbf{X}}$ Non-dimensional generalized displacement vector
$\dot{\tilde{\mathbf{X}}}$ Non-dimensional generalized velocity vector
$\ddot{\tilde{\mathbf{X}}}$ Non-dimensional generalized acceleration vector
x_θ Non-dimensional distance from elastic axis to center of gravity, positive for center of gravity aft of elastic axis, $x_\theta = \tilde{S}_\theta/(\tilde{m}b)$
z Spanwise coordinate (m)
$\bar{z}$ Non-dimensional spanwise coordinate, $\bar{z} = z/s$
α_{eff} Effective angle of attack, positive for clockwise about the elastic axis (EA), $\alpha_{\text{eff}} = \alpha_o - \theta$ (°)
α_{Lo} Angle of attack at which the lift force is zero, positive for clockwise about the elastic axis (EA) (°)
α_o Geometric angle of attack, positive for clockwise with respect to the incoming flow (°)
Θ Instantaneous twisting deformation positive for counter-clockwise about the elastic axis (EA) (°)
$\tilde{\theta}$ Spanwise twisting deformation positive for counter-clockwise about the elastic axis (EA) (°)
$\tilde{\theta}_{\text{elastic}}$ Aero/hydroelastic twist deformation (°)
λ Eigenvalue
μ Relative mass ratio: the ratio between structural inertial force and fluid inertial force for bending motion, $\mu = \tilde{m}/(\pi\rho_f b^2 s)$
ν_s Structural Poisson ratio
ν_f Fluid kinematic viscosity (m^2/s)
ρ_s Structural density (kg/m^3)
ρ_f Fluid density (kg/m^3)
$\zeta_{f,h}$ Fluid bending damping coefficient
$\zeta_{f,\theta}$ Fluid twisting damping coefficient
$\zeta_{s,h}$ Structural bending damping coefficient
$\zeta_{s,\theta}$ Structural twisting damping coefficient
ζ_T Total damping coefficient
$\zeta_{T,h}$ Total bending damping coefficient
$\zeta_{T,\theta}$ Total twisting damping coefficient
Ω Structural bending to twisting frequency ratio: the ratio between first in-air natural bending frequency and first in-air natural twisting frequency, $\Omega = \omega_h/\omega_\theta$

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ω_d Damped natural frequency, $\omega_d = 2\pi f_d$ (Hz)
ω_h First in-vacuum (or in-air) natural bending frequency, $\omega_h = 2\pi f_h$ (Hz)
ω_n Undamped natural frequency, $\omega_n = 2\pi f_n$ (Hz)
ω_θ First in-vacuum (or in-air) natural twisting frequency, $\omega_\theta = 2\pi f_\theta$ (Hz)

I. INTRODUCTION

Concurrent with smart material development, it is possible to take advantage of passive flexible body deformations to improve performance, efficiency, maneuverability, and harvest flow kinetic energy. An improved understanding of the aeroelastic or hydroelastic response and stability is needed to quantify the available operation space for the advanced design of the multi-functional flexible foils. A general understanding of the maximum deformation that can be achieved by flexible foils before material or instability failure is also required.

The general performances of flexible hydrofoils contrast to airfoils because water density is three orders of magnitude higher than air density.[1–3] The higher hydrodynamic loading forces of the hydrofoil have a much lower aspect ratio (AR) than their counterparts in air to avoid premature material failure. Since engineered hydrofoils typically have aspect ratios higher than one to maximize efficiency, the spanwise flexibility tends to dominate rather than the chordwise flexibility. The spanwise flexibility effects play an essential role in determining the aero/hydroelastic performances of efficiency.[4–7] Moreover, the maximum tip deformation of engineered hydrofoils tends to be limited to avoid excessive flow-induced vibrations, as well as material or instability failure.[2,8] The flow-induced bend-twist coupling terms on the hydrofoils affect the natural frequencies and damping coefficients, particularly for the twisting mode.[3] The natural frequencies of the flexible hydrofoils tend to be much lower than those of airfoils. In contrast, the damping coefficients of the flexible hydrofoils tend to be much higher than those of airfoils due to the significant contribution of the fluid added mass and hydrodynamic damping effects, which can vary with vibration mode, submergence, speed, cavitation, and ventilation.[2,3,9–12] In addition, theoretical fluid disturbing forces vary with the speed's square, and they are dependent on the flow directions, submergence, cavitation, and ventilation.[13–18] Therefore, it is important to understand the fluid effects on steady and dynamic responses of spanwise flexible foils due to the fluid-structure interaction (FSI), such as the limits of flexible deformations, aero/hydrodynamic performances, vibration characteristics, and the susceptibility to elastic mode instabilities. Parametric studies at the different fluids are also needed to avoid accelerated fatigue, noise, and vibration issues due to the excessive flow-induced vibrations or to take advantage of flow-induced vibrations to harvest energy.

This research aims to explore the influence of general critical geometric and material parameters on the steady and dynamic responses of spanwise flexible airfoils and hydrofoils. This research is limited to the rectangular cantilevered foils made of homogeneous isotropic linear elastic material and inviscid flow conditions. Specifically, we seek to compare (1) the influence of the relative magnitude of the solid and fluid terms between airfoils and hydrofoils, (2) the influence of the flow-induced bend-twist coupling terms between airfoils and hydrofoils, (3) the dependence of the vibration characteristics (natural frequencies and damping coefficients) with the solid-to-fluid mass ratio (representing different materials and fluid combinations) and velocity for both airfoils and hydrofoils, and (4) the maximum allowable elastic tip deformations for airfoils vs hydrofoils.

II. ANALYTIC MODEL

The analytical model uses a rectangular cantilevered foil with spanwise bending and twisting flexibilities only, which is a canonical proxy to more complex flexible lift generating devices (e.g., propellers and turbine blades, control surfaces, and wings). The material response is assumed to be isotropic and linear elastic;[10,19] the potential flow theory is assumed to be valid, and viscous effects will be discussed later in Sec. III C. Note that the anisotropic material responses will introduce material bend-twist coupling effects[12,19] and are outside the scope of this work. The foil is assumed to be deeply submerged such that the effects of the Froude number and cavitation number are negligible (see Young *et al.*[18] for a detailed discussion of ventilation and free surface effects).

For simplicity and validation purposes, a rectangular cantilevered NACA 0015 flexible foil is used, but significant conclusions can be applied to general foils. The foil structural response is simplified as a two-degrees of freedom (2-DOF) model characterized by the tip bending deformation (h) as a positive up direction and twisting deformations (θ) as a positive counterclockwise rotation (i.e., nose-down) shown in Fig. 1. The 2-DOF model can be derived by decomposing the generalized spanwise bending and twisting deformations, $\tilde{h}(\bar{z},\bar{t})$ and $\tilde{\theta}(\bar{z},\bar{t})$, as a function of the mode shapes in space and time,

$$\tilde{h}(\bar{z},\bar{t}) = h(\bar{t})f(\bar{z}), \tag{1}$$

$$\tilde{\theta}(\bar{z},\bar{t}) = \theta(\bar{t})g(\bar{z}), \tag{2}$$

where $\bar{z} = z/s$ is the non-dimensional spanwise coordinate originating from the root; s is the span of foil; $\bar{t}$ is the non-dimensional time; and $f(\bar{z})$ and $g(\bar{z})$ are the non-dimensional shape functions of the first bending and first twisting modes along the spanwise direction of a cantilevered beam with uniform cross section, as shown in Eqs. (3) and (4) and Fig. 2, where A is a constant number of 1.875,

$$f(\bar{z}) = \frac{\cosh(A\bar{z}) - \cos(A\bar{z}) - 0.734[\sinh(A\bar{z}) - \sin(A\bar{z})]}{2}, \tag{3}$$

$$g(\bar{z}) = \sin\left(\frac{\pi}{2}\bar{z}\right). \tag{4}$$

In Fig. 1, U is the free stream velocity as positive in the **X** direction; ab is the distance from the mid-chord to the elastic axis (EA), which is positive if EA is aft of the mid-chord; $c = 2b$ is the chord length, where b is the half of the chord length; $\tilde{K}_{s,h}(= \tilde{m}\omega_h^2)$ and $\tilde{K}_{s,\theta}(= \tilde{I}_\theta\omega_\theta^2)$ are the solid spanwise bending and twisting elastic restoring values, respectively; $\tilde{C}_{s,h}(= 2\tilde{m}\omega_h\zeta_{s,h})$ and $\tilde{C}_{s,\theta}(= 2\tilde{I}_\theta\omega_\theta\zeta_{s,\theta})$ are the solid spanwise bending and twisting damping values, respectively, where $\tilde{m}$ is the generalized solid mass; $\tilde{I}_\theta(= \tilde{m}r_\theta^2 b^2)$ is the generalized solid moment of inertia; r_θ is the radius of gyration about the EA; $\omega_h(= 2\pi f_h)$ and $\omega_\theta(= 2\pi f_\theta)$ are the in-air (or in-vacuum) natural bending and twisting frequencies, respectively; and $\zeta_{s,h}$ and $\zeta_{s,\theta}$ are the solid damping coefficients for bending and twisting motions, respectively.

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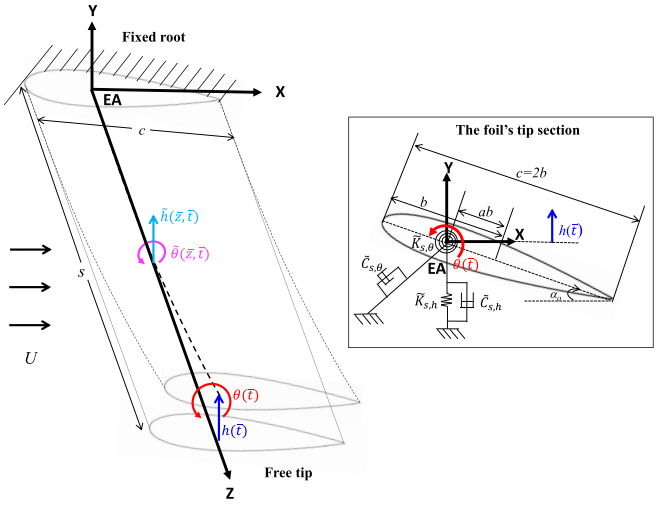


FIG. 1. A two-degree-of-freedom (2-DOF) model describing the spanwise bending and twisting flexibilities at the tip of a rectangular, cantilevered foil. Note that the springs and dampers are used to represent the foil spanwise flexibility and structural damping.

A. Governing equations

The governing equations use the non-dimensional generalized 2-DOF equations of motion (EOM) for the solid model and the non-dimensional inviscid Theodorson's fluid forces[20] for the fluid model. All dimensional parameters are consistently non-dimensionalized using the same characteristic length, mass, and time parameters, as shown in Table I.

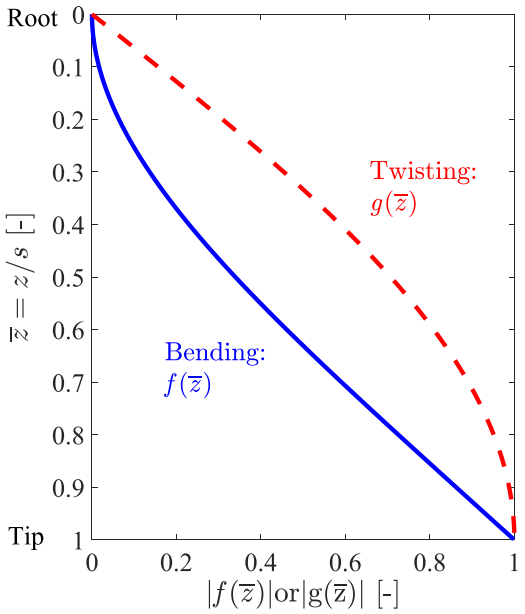


FIG. 2. Non-dimensional shape functions for bending and twisting motions, $f(\bar{z})$ and $g(\bar{z})$.

The non-dimensional generalized 2-DOF model in Eq. (5) governs the spanwise bending and twisting motions in the absence of external excitation force. The generalized formulation was derived by integrating the sectional solid and fluid forces along the spanwise direction obtained by applying the virtual work principle,[21,22]

$$\tilde{\mathbf{M}}_s \ddot{\tilde{\mathbf{X}}} + \tilde{\mathbf{C}}_s.\dot{\tilde{\mathbf{X}}} + \tilde{\mathbf{K}}_s\tilde{\mathbf{X}} = \tilde{\mathbf{F}}_{\text{fluid}}, \tag{5}$$

where $\tilde{\mathbf{X}} = [\tilde{h}/b, \tilde{\theta}]^\top$, $\dot{\tilde{\mathbf{X}}}$, and $\ddot{\tilde{\mathbf{X}}}$ are the non-dimensional generalized deformation, velocity, and acceleration vectors for bending and twisting motions at the EA of the foil, respectively; $\tilde{\mathbf{M}}_s$, $\tilde{\mathbf{C}}_s$, and $\tilde{\mathbf{K}}_s$ are the non-dimensional generalized solid inertial, damping, and restoring matrices, as given in Eqs. (6)–(8); and $\tilde{\mathbf{F}}_{\text{fluid}}$ is the non-dimensional generalized inviscid fluid force acting at the EA of the foil, as defined in Eq. (12),

$$\tilde{\mathbf{M}}_s = \begin{bmatrix} \mu S_{ff} & \mu x_\theta S_{fg} \\ \mu x_\theta S_{fg} & \mu r_\theta^2 S_{gg} \end{bmatrix}, \tag{6}$$

$$\tilde{\mathbf{C}}_s = \begin{bmatrix} 2\mu\Omega\zeta_{s,h} S_{ff} & 0 \\ 0 & 2\mu r_\theta^2 \zeta_{s,\theta} S_{gg} \end{bmatrix}, \tag{7}$$

$$\tilde{\mathbf{K}}_s = \begin{bmatrix} \mu\Omega^2 S_{ff} & 0 \\ 0 & \mu r_\theta^2 S_{gg} \end{bmatrix}, \tag{8}$$

where $\mu = \tilde{m}/(\pi\rho_f b^2 s)$ is the solid-to-fluid mass ratio; $x_\theta b$ is the distance from the EA to the center of gravity (CG), which is positive if CG is aft of EA; and $\Omega = \omega_h/\omega_\theta$ is the ratio between the in-air natural bending frequency (ω_h) and twisting frequency (ω_θ). The integrated shape functions of pure or combined bending and twisting motions can be defined as S_f, S_{ff}, S_g, S_{gg}, and S_{fg} in the following equations:

$$S_f = \int_0^1 f(\bar{z})d\bar{z}, \quad S_{ff} = \int_0^1 f^2(\bar{z})d\bar{z}, \tag{9}$$

$$S_g = \int_0^1 g(\bar{z})d\bar{z}, \quad S_{gg} = \int_0^1 g^2(\bar{z})d\bar{z}, \tag{10}$$

$$S_{fg} = \int_0^1 f(\bar{z})g(\bar{z})d\bar{z}. \tag{11}$$

The non-dimensional generalized inviscid fluid force ($\tilde{\mathbf{F}}_{\text{fluid}}$) is decomposed into steady ($\tilde{\mathbf{F}}_{\text{steady}}$) and dynamic ($\tilde{\mathbf{F}}_{\text{dynamic}}$) parts, and it is computed based on potential flow (i.e., inviscid, incompressible, and irrotational flows) without separation and stall,

$$\tilde{\mathbf{F}}_{\text{fluid}} = \tilde{\mathbf{F}}_{\text{steady}} + \tilde{\mathbf{F}}_{\text{dynamic}}. \tag{12}$$

$\tilde{\mathbf{F}}_{\text{steady}}$ is the generalized steady fluid force on the rigid foil at the steady flow condition, which is calculated via the thin airfoil theory[23] with the small deformation assumption shown in Eq. (13). $\tilde{\mathbf{F}}_{\text{steady}} = [\tilde{\text{L}}_{\text{steady}}, \tilde{\text{M}}_{\text{steady}}]^\top$ is decomposed into the lift and moment acting at the aerodynamic center (AC) due to the nonsymmetrical foil geometry and the initial alignment of the foil with respect to the inflow, at which the both bending and twisting restoring forces are zero,

$$\tilde{\mathbf{F}}_{\text{steady}} = \begin{bmatrix} \tilde{\text{L}}_{\text{steady}}, \\ \tilde{\text{M}}_{\text{steady}} \end{bmatrix} = \frac{\bar{U}^2}{\pi} \begin{bmatrix} C_{L_\alpha}(\alpha_o - \alpha_{Lo}) S_f \\ 2C_{M_{AC}} S_g - C_{L_\alpha}(\alpha_o - \alpha_{Lo}) e S_g \end{bmatrix}, \tag{13}$$

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TABLE I. Characteristic parameter. ρ_f is the fluid density.

Variable	Unit	Characteristic parameter
Length	m	b
Mass	kg	$\pi\rho_f b^2 s$
Time	s	$1/\omega_\theta$

where $\bar{U} = U/(\omega_\theta b)$ is the reduced velocity; α_o is the initial geometric angle of attack defined as a positive clockwise rotation; α_{Lo} is the zero lift angle of attack, i.e., the value of angle of attack when lift equals zero; $C_{M_{AC}}$ is the moment coefficient at the aerodynamic center (AC); $e\left(= a + \frac{1}{2}\right)$ is the non-dimensional distance from EA to the theoretical AC; and C_{L_α} is the three-dimensional (3D) lift slope for the foil, which can be obtained from the following equation:

$$C_{L_\alpha} = \frac{dC_L}{d\alpha} = \frac{a_o}{1 + \frac{a_o}{\pi e_o \mathrm{AR}}}, \tag{14}$$

where $a_o = 2\pi$ is the lift slope for the foil's section; $e_o = 0.8$ is the Oswald efficiency factor for the conventional aircraft, the 3D correction for the cross-flow or spanwise effects and AR is the aspect ratio of the wing, $\mathrm{AR} = 2s/c$, for a cantilevered wing to account for the image effect at the foil root. Note that α_{Lo} and $C_{M_{AC}}$ are both zero for a symmetric foil with no camber.

The generalized steady lift and moment coefficients[24] ($\tilde{C}_{\mathrm{L,steady}}$ and $\tilde{C}_{\mathrm{M,steady}}$) are expressed in the following equations:

$$\tilde{C}_{\mathrm{L,steady}} = C_{L_\alpha}(\alpha_o - \alpha_{Lo})S_f, \tag{15}$$

$$\tilde{C}_{\mathrm{M,steady}} = \left[C_{M_{AC}} - \frac{C_{L_\alpha}}{2}(\alpha_o - \alpha_{Lo})e\right]S_g. \tag{16}$$

$\tilde{\mathbf{F}}_{\mathrm{dynamic}} = [\tilde{\mathrm{L}}_{\mathrm{dynamic}}, \tilde{\mathrm{M}}_{\mathrm{dynamic}}]^\top$ is the generalized dynamic fluid force at the unsteady flow condition using the Theodorsen's approach,[20] which is expressed in terms of inviscid fluid dynamic lift and moment acting on the EA at the foil, as shown in Eq. (17). It should be noted that the Theodorsen's approach assumes that the flow is always attached, and the foil is thin, has a small initial angle of attack, and undergoes small harmonic bending and twisting motions. The dynamic fluid force is composed of both circulatory and non-circulatory terms. The non-circulatory terms are from the pressure difference between the top and bottom foil surfaces as well as the fluid added mass terms. The circulatory terms are from the pressure difference caused by the velocity induced by vortices on the wing and wakes,

$$\tilde{\mathbf{F}}_{\mathrm{dynamic}} = -\tilde{\mathbf{M}}_f \ddot{\tilde{\mathbf{X}}} - \tilde{\mathbf{C}}_f \dot{\tilde{\mathbf{X}}} - \tilde{\mathbf{K}}_f \tilde{\mathbf{X}}, \tag{17}$$

where $\tilde{\mathbf{M}}_f$, $\tilde{\mathbf{C}}_f$, and $\tilde{\mathbf{K}}_f$ are the non-dimensional generalized fluid inertial, damping, and disturbing force matrices, respectively, as given in the following equations:

$$\tilde{\mathbf{M}}_f = \begin{bmatrix} S_{ff} & -aS_{fg} \\ -aS_{fg} & \left(\frac{1}{8} + a^2\right)S_{gg} \end{bmatrix}, \tag{18}$$

$$\tilde{\mathbf{C}}_f = \bar{U}\begin{bmatrix} 2C(k)S_{ff} & (1 - 2C(k)d)S_{fg} \\ -2C(k)eS_{fg} & d[2C(k)e - 1]S_{gg} \end{bmatrix}, \tag{19}$$

$$\tilde{\mathbf{K}}_f = \bar{U}^2\begin{bmatrix} 0 & 2C(k)S_{fg} \\ 0 & -2C(k)eS_{gg} \end{bmatrix}, \tag{20}$$

where $d\left(= a - \frac{1}{2}\right)$ is the non-dimensional distance from the EA to the three-quarter-chord length ($3c/4$); $k(= \omega b/U)$ is the reduced frequency of the oscillating foil, where $\omega(= 2\pi f)$ is the oscillation frequency; and $C(k)$ is the Theodorsen's circulation function, which accounts for the circulatory load on the foil induced by the shed vortices in the wake, as defined in the following equation:

$$C(k) = \frac{H_1^2(k)}{H_1^2(k) + iH_0^2(k)}, \tag{21}$$

where $H_1^2(k)$ and $H_0^2(k)$ are the Hänkel functions (i.e., Bessel functions of the third kind). It should be noted that $C(k)$ is predicted via an iterative process.

For the flexible foil, the total angle of attack is taken as the sum of the initial geometry angle of attack (α_o), zero lift angle of attack (α_{Lo}), and aero/hydroelastic twist deformation ($\tilde{\theta}_{\mathrm{elastic}}$), which is an additional increment due to the elastic twist of the spring from the inviscid fluid force using the Theodorsen's approach[20] at the unsteady flow condition, as shown in the following equation:

$$\alpha = \alpha_o - \alpha_{Lo} - \tilde{\theta}_{\mathrm{elastic}}. \tag{22}$$

The generalized aero/hydroelastic lift and moment coefficients ($\tilde{C}_{\mathrm{L,elastic}}$ and $\tilde{C}_{\mathrm{M,elastic}}$)[24] of the flexible foil are related to the total angle of attack and are expressed in the following equations:

$$\tilde{C}_{\mathrm{L,elastic}} = C_{L_\alpha}(\alpha_o - \alpha_{Lo} - \tilde{\theta}_{\mathrm{elastic}})S_f, \tag{23}$$

$$\tilde{C}_{\mathrm{M,elastic}} = \left[C_{M_{AC}} - C_{L_\alpha}\frac{e}{2}(\alpha_o - \alpha_{Lo} - \tilde{\theta}_{\mathrm{elastic}})\right]S_g. \tag{24}$$

The inviscid time-domain (TD) fluid-structure interaction (FSI) response of the flexible foil was modeled by using the fully coupled (FC) method, as shown in Eq. (25). The inviscid TD FSI model is coupled with an inviscid Theodorsen's approach for the fluid solver and a two-degree-off-freedom (2-DOF) generalized equation of motion (EOM) for the solid solver shown in Fig. 1,

$$\tilde{\mathbf{M}}_s(\ddot{\tilde{\mathbf{X}}})_n + \tilde{\mathbf{C}}_s(\dot{\tilde{\mathbf{X}}})_n + \tilde{\mathbf{K}}_s(\tilde{\mathbf{X}})_n - (\tilde{\mathbf{F}}_{\mathrm{dynamic}})_n = (\tilde{\mathbf{F}}_{\mathrm{steady}})_n, \tag{25}$$

where the subscript n denotes the discrete time-level using the Crank–Nicolson time integration method; the Theodorsen's circulation function $C(k)$ in $\mathbf{F}_{\mathrm{dynamic}}$, as a function of the reduced frequency (k) in Eqs. (19) and (20), is predicted via an iterative process in Eq. (25); and the tip bending and twisting deformations ($\tilde{h}/b$, $\tilde{\theta}$) of the inviscid TD FSI model are computed in $\tilde{\mathbf{X}}$ once we solve the motions.

B. Non-dimensional critical parameters

For the sake of simplicity, inviscid analyses [e.g., inviscid theory, inviscid frequency-domain (FD) FSI simulation, and inviscid time-domain (TD) FSI simulation] are conducted to examine a large number of parametric spaces in order to study the influence of material, geometric, and flow parameters. The non-dimensional parameters

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TABLE II. Non-dimensional parameters of the flexible foils. ν_f is the fluid kinematic viscosity and $\tilde{S}_\theta = \tilde{m}x_\theta b$ is the generalized static imbalance.

Parameter	Symbol	Definition	Physical meaning
Solid-to-fluid mass ratio	μ	$\frac{\tilde{m}}{\pi\rho_f b^2 s}$	Ratio between the foil inertial force and fluid inertial force for bending motion
Solid bending-to-twisting frequency ratio	Ω	$\frac{\omega_h}{\omega_\theta}$	Ratio between the in-air foil natural bending and twisting frequency
Solid damping coefficient	$\zeta_{s,h}$, $\zeta_{s,\theta}$	$\frac{\tilde{C}_{s,h}}{2\tilde{m}\omega_h}$, $\frac{\tilde{C}_{s,\theta}}{2\tilde{I}_\theta\omega_\theta}$	Foil structural bending and twisting damping coefficients
Elastic axis	a	$\cdots$	Non-dimensional distance from the mid-chord to foil elastic axis (EA), positive if EA is aft of the mid-chord
Center of pressure	e	$e = a + \frac{1}{2}$	Non-dimensional distance from EA to the theoretical aerodynamic center (AC)
Center of gravity	x_θ	$\frac{\tilde{S}_\theta}{\tilde{m}b}$	Non-dimensional distance from EA to the foil center of gravity (CG), positive for CG aft of EA
Radius of gyration	r_θ	$\sqrt{\frac{\tilde{I}_\theta}{\tilde{m}b^2}}$	Non-dimensional radius of gyration about EA
Aspect ratio	AR	$\frac{2s}{c}$	Effective span to chord ratio for a cantilever foil (Note that the factor of 2 is used to account for the wall image effect.)
Reynolds number	Re	$\frac{Uc}{\nu_f}$	Ratio between the fluid inertial force and fluid viscous force
Reduced velocity	$\bar{U}$	$\frac{U}{\omega_\theta b}$	Ratio between the fluid convection frequency and foil in-air natural twisting frequency
Reduced frequency	k	$\frac{\omega b}{U}$	Ratio between the foil oscillation frequency and fluid convection frequency

governing the steady and dynamic responses of the flexible foils are listed in Table II. The non-dimensional critical parameters of the natural frequencies, fluid damping coefficients, and aero/hydro-elastic deformations with varying the solid-to-fluid mass ratio, velocity ratio, and aspect ratio drive the critical design parameters of the flexible foil, as shown in Sec. II C.

Since the stability of flexible foils depends on the flow velocity, the reduced *stable* velocity ratio ($\bar{U}_{\text{stable}}$) is defined in Eq. (26), which is the velocity ratio between the free-stream (U) and the divergence speed (U_d),

$$\bar{U}_{\text{stable}} = \frac{U}{U_d} = \bar{U}\sqrt{\frac{2C(k)e}{\mu r_\theta^2}}, \tag{26}$$

where $\bar{U} = U/(\omega_\theta b)$ is the reduced velocity. It should be noted that the divergence corresponds to the critical speed at which the fluid disturbing moment is equal to the flexible foil elastic restoring moment, leading to zero effective twisting stiffness, and hence unbounded deformation with slight flow-induced perturbations. The divergence is dependent on the solid-to-fluid mass ratio because the fluid disturbing force is proportional to the fluid density, and hence, it is a higher value in the airfoil than the hydrofoil.

From the inviscid theory, the non-dimensional, *uncoupled mode* (UM), bending and twisting natural frequencies ($\bar{f}_{h-\text{UM}}$ and $\bar{f}_{\theta-\text{UM}}$), as well as fluid bending and twisting damping coefficients ($\zeta_{f,h-\text{UM}}$ and $\zeta_{f,\theta-\text{UM}}$) in Sec. III are obtained by neglecting the flow-induced bend-twist coupling terms, i.e., the off diagonal terms in Eqs. (18)–(20) given by Blake and Maga,[25] and are found as

$$\bar{f}_{h-\text{UM}} = \frac{f^*_{h-\text{UM}}}{f_\theta} = \Omega\sqrt{\frac{\mu}{\mu+1}}, \tag{27}$$

$$\bar{f}_{\theta-\text{UM}} = \frac{f^*_{\theta-\text{UM}}}{f_\theta} = \sqrt{\frac{\mu r_\theta^2 - \mu r_\theta^2 \bar{U}^2_{\text{stable}}}{\mu r_\theta^2 + \left(\frac{1}{8} + a^2\right)}}, \tag{28}$$

$$\zeta_{f,h-\text{UM}} = \frac{r_\theta\sqrt{C(k)}\bar{U}_{\text{stable}}}{\Omega\sqrt{2e(\mu+1)(1+2a)}}, \tag{29}$$

$$\zeta_{f,\theta-\text{UM}} = \frac{d[2C(k)e-1]\bar{U}_{\text{stable}}}{\sqrt{8eC(k)(1-\bar{U}^2_{\text{stable}})\left[\mu r_\theta^2 + \left(\frac{1}{8}+a^2\right)\right]}}, \tag{30}$$

where the superscript "$*$" refers to the in-water frequency. It should be noted that the inviscid theory approach in Eqs. (27)–(30) is simple to estimate the natural frequencies and fluid damping coefficients by ignoring the flow-induced bend-twist coupling terms. However, the influence of flow-induced bend-twist coupling is expected to be relatively more important as the flow speed increases. The *coupled mode* (CM) solutions, including the flow-induced bend-twist coupling

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terms, i.e., the off diagonal terms in Eqs. (18)–(20), are necessary to accurately predict the natural frequencies and fluid damping coefficients. Hence, the inviscid FD FSI model and the inviscid TD FSI model are considered.

The inviscid FD FSI solution assumes simple harmonic deformations of $\tilde{\mathbf{X}}_{\mathrm{FD}}$ in Eq. (5) for the eigenvalue problem,

$$\tilde{\mathbf{X}}_{\mathrm{FD}} = \mathbf{A}e^{\lambda t}, \tag{31}$$

where $\mathbf{A}$ is a constant vector that depends on the initial bending and twisting deformations, and λ is the eigenvalue of Eq. (5) expressed by the real, Real(λ), and imaginary, Imag(λ) components, as shown in the following equation:

$$\lambda = -\zeta_{T_{\mathrm{FD}}}\omega_n \pm i\,\omega_{n_{\mathrm{FD}}}\sqrt{1-\zeta_{T_{\mathrm{FD}}}^2}, \tag{32}$$

where ζ_T is the total damping coefficient; $\omega_{n_{\mathrm{FD}}}$ is the undamped natural frequency; and $\omega_{d_{\mathrm{FD}}}$ is the damped natural frequency, as given in the following equations:

$$\zeta_{T_{\mathrm{FD}}} == \frac{-\mathrm{Real}(\lambda)}{\sqrt{\mathrm{Real}(\lambda)^2+\mathrm{Imag}(\lambda)^2}}, \tag{33}$$

$$\omega_{n_{\mathrm{FD}}} = 2\pi f_{n_{\mathrm{FD}}} = \sqrt{\mathrm{Real}(\lambda)^2+\mathrm{Imag}(\lambda)^2}, \tag{34}$$

$$\omega_{d_{\mathrm{FD}}} = 2\pi f_{d_{\mathrm{FD}}} = \omega_{n_{\mathrm{FD}}}\sqrt{1-\zeta_{T_{\mathrm{FD}}}^2}. \tag{35}$$

It should be noted that both the inviscid linear theory and the inviscid FD FSI simulation assume the simple harmonic motion. However, the inviscid FD FSI solution can be different from the inviscid linear theory due to the inclusion of the flow-induced bend-twist coupling terms, i.e., the off diagonal terms in Eqs. (18)–(20).

The inviscid TD FSI simulation solves the motion from Eq. (25), and the natural bending and twisting frequencies are predicted by the fast Fourier transform (FFT) method. The total damping coefficients ($\zeta_{T_{\mathrm{TD}}}$) in Eq. (36) are calculated by the mobility peak power down method,[26]

$$\zeta_{T_{\mathrm{TD}}} = \frac{f_2 - f_1}{2f_i\sqrt{p-1}}, \tag{36}$$

where p is the power down point; f_i is the peak frequency; and f_1 and f_2 correspond to the frequencies on the left and right positions at pth mobility peak amplitude, i.e., the peak amplitude of f_i divided by the square root of p.

The limit of the aero/hydro-elastic bending and twisting deformations on the flexible foil at the unsteady flow condition is critical to avoid structural failure. From the equation of moment equilibrium, the aero/hydro-elastic bending and twisting deformations of the flexible foil are defined in Eqs. (37) and (38), i.e., all applied moments about the elastic axis must equal the restoring moment, which is the torsional reaction of the elastic support due to the inviscid fluid force[20] at the unsteady flow condition,

$$\tilde{\theta}_{\mathrm{elastic}} = \frac{\bar{U}^2_{\mathrm{stable}}\left[2C_{M,AC}S_g - C_{L_\alpha}e(\alpha_o-\alpha_{Lo})S_f\right]}{\left[(1+2a)-2\bar{U}^2_{\mathrm{stable}}e\right]\pi C(k)S_{gg}}, \tag{37}$$

$$\frac{\tilde{h}_{\mathrm{elastic}}}{b} = \frac{\bar{U}^2_{\mathrm{stable}}r_\theta^2\left[C_{L_\alpha}(\alpha_o-\alpha_{Lo})S_f - 2\pi C(k)e\tilde{\theta}_{\mathrm{elastic}}S_{fg}\right]}{(1+2a)C(k)\pi\Omega^2 S_{ff}}. \tag{38}$$

C. Model parameters

Analytical inviscid simulations are performed on NACA 0015 airfoils and hydrofoils made of different materials [e.g., polyacetate (POM), aluminum (Al), and stainless steel (SS)]. Table III lists the range of model parameters and fluid properties for rectangular cantilevered NACA 0015 flexible foils with identical geometries but made of different materials. For all the cases, the water is assumed for the fluid density of $\rho_f = 1000\,\mathrm{kg/m^3}$ and fluid kinematic viscosity of $\nu_f = 1\times10^{-6}\,\mathrm{m^2/s}$; the air is assumed for $\rho_f = 1.185\,\mathrm{kg/m^3}$ and $\nu_f = 1.55\times10^{-5}\,\mathrm{m^2/s}$; and the foil is assumed to be initially at rest. The subscript s and f denote the solid and fluid parameters, respectively; ρ_s is the solid density; E_s is the Young's modulus; and ν_s is the Poisson ratio. Note that the solid damping coefficients ($\zeta_{s,h}$ and $\zeta_{s,\theta}$) in Table III are assumed values based on measurements reported in the literature. As will be shown later in Sec. III C, the fluid damping coefficient of hydrofoils is typically more than an order of magnitude higher than the solid damping coefficient, and hence, the solid damping coefficient tends to be inconsequential for hydrofoils. The in-air natural bending and twisting frequencies (f_h and f_θ) were calculated following the theoretical formulas[27] given in Eqs. (39) and (40), which were derived for rectangular, homogeneous, isotropic, and cantilevered plates,

$$\omega_h = 2\pi f_h = \frac{\beta_1}{s}\sqrt{\frac{E_s\left(\frac{y_t}{s}\right)^2}{12(1-\nu_s^2)\rho_s}}, \tag{39}$$

$$\omega_\theta = 2\pi f_\theta = \frac{\beta_2}{s}\sqrt{\frac{E_s\left(\frac{y_t}{s}\right)^2}{12(1-\nu_s^2)\rho_s}}, \tag{40}$$

where $\beta_1 = 2.88$ and $\beta_2 = 13.84$ were determined by matching the in-air natural frequencies of the NACA 0015 POM hydrofoil measured at the French Naval Academy Research Institute (IRENav).[2]

III. INVISCID PARAMETRIC STUDIES

The inviscid parametric studies are considered in this section via the inviscid theory (*uncoupled mode*, UM), inviscid frequency-domain

TABLE III. Model parameters for NACA 0015 airfoils and hydrofoils made of different materials: polyacetate (POM), aluminum (Al), and stainless steel (SS).

Parameter		Unit	POM	Al	SS
ρ_s		$\mathrm{kg/m^3}$	1420	2700	8010
E_s		GPa	2.9	69	203
$\sigma_{s,ut}$		GPa	0.07	0.29	0.86
f_h		Hz	81	284	274
f_θ		Hz	390	1374	1324
$\Omega = f_h/f_\theta$		⋯	0.21	0.21	0.21
ν_s		⋯	0.35	0.33	0.3
$\zeta_{s,h}$, $\zeta_{s,\theta}$		⋯	0.02	0.002	0.001
Hydrofoil	ρ_s/ρ_f	⋯	1.42	2.7	8.01
	$\sqrt{\mu}$	⋯	0.43	0.59	1.02
Airfoil	ρ_s/ρ_f	⋯	1.2×10^3	2.28×10^3	6.76×10^3
	$\sqrt{\mu}$	⋯	12.53	17.28	29.76

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(FD) FSI simulation, and inviscid time-domain (TD) FSI simulation. The governing non-dimensional material, geometric, and flow parameters are systematically varied to examine the steady and dynamic responses of the flexible airfoils vs hydrofoils. In particular, the relative magnitude of the solid and fluid terms, natural frequencies, fluid damping coefficients, and elastic deformations with varying solid-to-fluid mass ratios and velocity are examined. The inviscid FSI models with and without the flow-induced bend-twist coupling terms are investigated. It should be noted that the flow-induced bend-twist coupling or off diagonal terms in Eqs. (18)–(20) from the Theodorson's fluid forces[20] arise from the center of the pressure being away from the elastic axis, leading to a twisting moment, and circulatory $C(k)$ terms caused by the shed vortices in the wake. Therefore, the flow-induced bend-twist coupling terms of the fluid inertial, damping, and disturbing force terms affect the natural frequencies and total damping coefficients. The flow-induced bend-twist coupling terms are important (particularly for the twisting mode) for the cases with low solid-to-fluid added mass ratios ($\mu \leq 1$), and their relative importance increases with higher inflow velocity and lower solid-to-fluid added mass ratio.[3]

Validation studies of the inviscid TD FSI predictions were obtained using the FC method with experimental measurements of rigid and flexible hydrofoils in both elastic and dynamic conditions, which can be found in previous works.[1–3,10,28] Experimental and numerical investigations of viscous effects have been discussed in previous works.[1–3,10,21,28] Good agreements were observed between measurements and predictions for aero/hydrodynamic load coefficients, vibration frequencies, damping characteristics, and flow-induced vibrations of cantilevered flexible air/hydrofoils in steady/unsteady flows.

A. Influence of the relative magnitude of the solid and fluid terms

To see the influence of relative solid and fluid terms, Fig. 3 shows the relative magnitude between the coefficients for each of the solid and fluid terms using the order of magnitude method in Eqs. (41) and (42) for POM hydrofoil ($\sqrt{\mu} = 0.43$), SS hydrofoil ($\sqrt{\mu} = 1.02$), POM airfoil ($\sqrt{\mu} = 12.53$), and SS airfoil ($\sqrt{\mu} = 29.76$),

$$\left|\frac{\tilde{\mathbf{M}}_s}{\tilde{\mathbf{M}}_s}\right| = 1, \quad \left|\frac{\tilde{\mathbf{C}}_s}{\tilde{\mathbf{M}}_s}\right| \propto \Omega, \quad \left|\frac{\tilde{\mathbf{K}}_s}{\tilde{\mathbf{M}}_s}\right| \propto \Omega^2, \tag{41}$$

$$\left|\frac{\tilde{\mathbf{M}}_f}{\tilde{\mathbf{M}}_s}\right| \propto \frac{1}{\mu}, \quad \left|\frac{\tilde{\mathbf{C}}_f}{\tilde{\mathbf{M}}_s}\right| \propto \frac{\bar{U}_{\text{stable}}}{\mu}, \quad \left|\frac{\tilde{\mathbf{K}}_f}{\tilde{\mathbf{M}}_s}\right| \propto \frac{\bar{U}^2_{\text{stable}}}{\mu}, \tag{42}$$

where $\Omega = 0.21$ is fixed for all cases in this work.

The results in Fig. 3 show that for low mass ratio cases ($\sqrt{\mu} \ll 1$, e.g., POM hydrofoil), fluid inertial terms are much more significant than solid inertial terms; while for high mass ratio cases ($\sqrt{\mu} \gg 1$, e.g., POM airfoil), solid inertial terms are much higher than fluid inertial terms. Note that all the fluid forces are proportional to ρ_f, and hence the $\frac{1}{\mu}$ dependency in Eq. (42), which is responsible for the higher fluid-to-solid force ratios. As already shown in Eqs. (19) and (20) and in Eq. (42), the (linear) fluid (radiation) damping term is proportional to the velocity, and the (linear) fluid disturbing term is proportional to the velocity square. As the velocity increases, although the high fluid damping will tend to damp out the vibrations, the foil may become increasingly susceptible to static divergence, as the fluid disturbing force/moment terms increase with velocity square. To ensure stability, the solid stiffness force and moment terms need to be sufficiently higher than the fluid stiffness, where the latter have negative values as they represent the fluid disturbing force and moment. The stability requirement is related to the foil restoring moment, which is related to the material shear modulus.

To see the dynamic response of the relative solid and fluid force coefficients, Fig. 4 shows the time-histories of lift and moment coefficients of the inviscid TD FSI model including the flow-induced bend-twist coupling terms, i.e., coupled mode (CM), on the cantilevered NACA 0015 POM hydrofoil ($\sqrt{\mu} = 0.43$) and POM airfoil ($\sqrt{\mu} = 12.53$) at $\bar{U}_{\text{stable}} = 0.2$, AR = 4, and $\alpha_o = 8°$. The subscripts $_{\mathbf{M}}$, $_{\mathbf{C}}$, and $_{\mathbf{K}}$ denote the force coefficients induced by the inertial, damping, and stiffness terms, respectively; and the subscripts $_{\mathbf{s}}$ and $_{\mathbf{f}}$ denote the solid and fluid terms, respectively. The results show that lift and moment coefficients induced by the solid stiffness term are dominant both on the airfoil and hydrofoil since $\bar{U}_{\text{stable}} = 0.2$, which is necessary to ensure stability. The lift and moment coefficients induced by the fluid inertia and fluid damping terms are dominant on the hydrofoil because of the high fluid density. In contrast, the lift and moment coefficients induced by the solid inertia term is dominant on the airfoil. Hence, the lift and moment load fluctuations associated with flow-induced vibrations damp out much faster with the lower amplitude on the hydrofoil than airfoil shown in Fig. 5.

B. Influence of the flow-induced bend-twist coupling

For the influence of the flow-induced bend-twist coupling, the inviscid TD FSI model is considered by either including the flow-induced bend-twist coupling terms, i.e., the off diagonal terms in Eqs. (18)–(20), shown as *coupled mode* (CM) or excluding them shown as *uncoupled mode* (UM). Figures 5 and 6 show the time-histories of the tip bending and twisting deformations as well as the lift and moment coefficients of the inviscid TD FSI model for a cantilevered NACA 0015 POM hydrofoil ($\sqrt{\mu} = 0.43$) and POM airfoil ($\sqrt{\mu} = 12.53$) at $\bar{U}_{\text{stable}} = 0.2$, AR = 4, and $\alpha_o = 8°$. The subscripts $_{\mathbf{M}}$, $_{\mathbf{C}}$, and $_{\mathbf{K}}$ denote the force coefficients induced by the inertial, damping, and stiffness terms, respectively, and the subscripts $_{\mathbf{s}}$ and $_{\mathbf{f}}$ denote the solid and fluid terms.

The results in Figs. 5 and 6 show that the flow-induced bend-twist coupling effect is negligible on the airfoil compared to the hydrofoil due to the low fluid density. However, ignoring the flow-induced bend-twist coupling effect leads to significant differences in the twisting response and moment coefficient for the hydrofoil. In particular, the inviscid TD FSI, UM, model of the hydrofoil over-predicted the fluid damping, particularly for the twisting mode, which leads to under-prediction of the twist and moment coefficient fluctuation amplitudes; hence, the fluctuations damp out much faster with the lower amplitude.

C. Influence of the solid-to-fluid mass ratio and velocity

To see the influence of the solid-to-fluid mass ratio and velocity, the measured and predicted non-dimensional bending and twisting natural frequencies ($\overline{f_h} = f_h^*/f_\theta$ and $\overline{f_\theta} = f_\theta^*/f_\theta$) as well as total bending and twisting damping coefficients ($\zeta_{T,h}$ and $\zeta_{T,\theta}$) are plotted in

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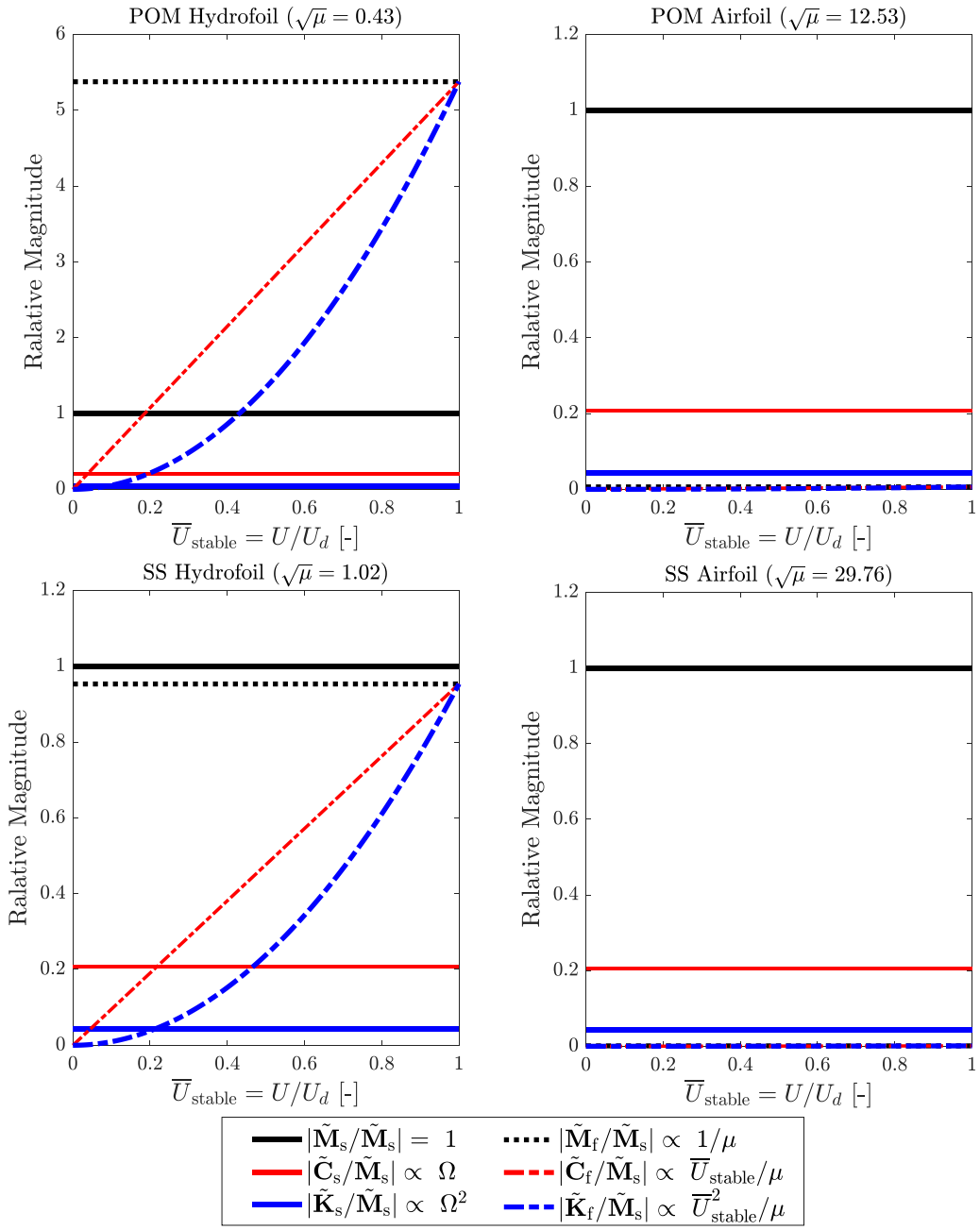


FIG. 3. The relative magnitude between the coefficients for each of the solid and fluid terms using the order of magnitude method for POM hydrofoil, POM airfoil, SS hydrofoil, and SS airfoil.

Figs. 7 and 8 on the cantilevered foils made of different materials derived from the flow-induced spanwise tip bending and twisting velocity fluctuations with varying $\bar{U}_{\text{stable}}$ at AR $= 4$ and $\alpha_o = 8°$.

The model parameters of airfoils and hydrofoils (i.e., operating in air and water) made of different materials [e.g., polyacetate (POM), aluminum (Al), and stainless steel (SS)] are given in Table III. In Figs. 7 and 8, the inviscid theory results exclude the flow-induced bend-twist coupling terms, i.e., uncoupled mode (UM), that are calculated using Eqs. (27)–(30) shown by dashed-dotted lines; the inviscid frequency-domain (FD) FSI results include the flow-induced bend-twist coupling terms, i.e., coupled mode (CM), that are solved by eigenvalue problems and given in Eqs. (33)–(35) and are shown by cross symbols; the inviscid time-domain (TD) FSI results include the flow-induced bend-twist coupling terms,

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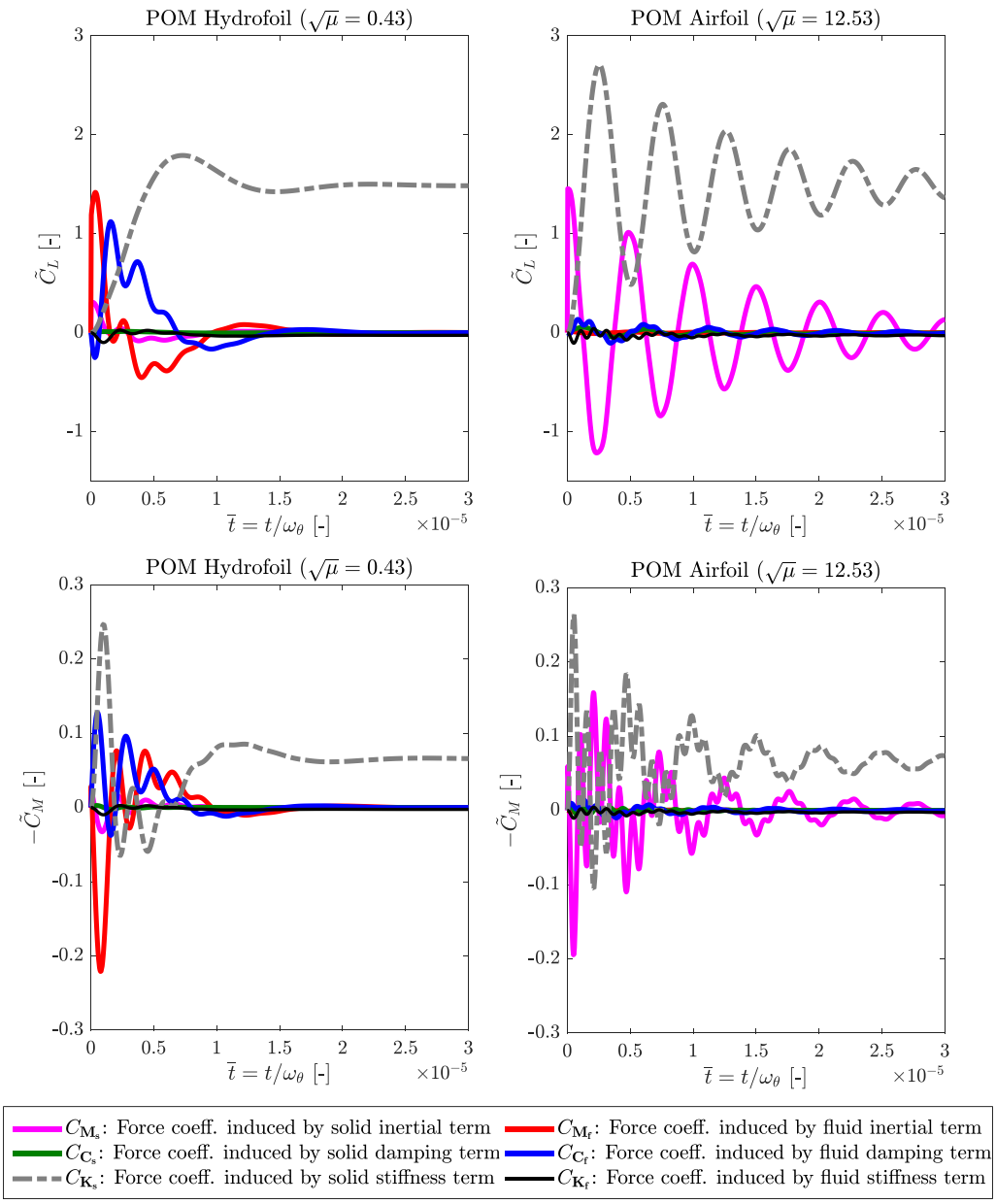


FIG. 4. The time-histories of the fluid and solid inertial, damping, and stiffness components of the lift coefficients (top row) and moment coefficients (bottom row). Results are obtained using the inviscid TD FSI model on the cantilevered NACA 0015 POM hydrofoil and POM airfoil at $\bar{U}_{\text{stable}} = 0.2$, AR = 4, and $\alpha_o = 8^\circ$.

i.e., coupled mode (CM), that are derived by taking the fast Fourier transform (FFT) response of the time history from the solution of Eq. (25) and are shown by open symbols; and the experimental results[2,25,29] with three different materials are shown by filled symbols.

Chae *et al.*'s experimental results[2] are obtained from measurements of the natural flow-induced vibrations of rectangular, cantilevered flexible hydrofoils. The experimental measurements were conducted for the cantilevered polyacetate (POM) NACA 0015 hydrofoil with an effective aspect ratio of 4 and were carried out in a circulating water tunnel at the French Naval Academy Research Institute (IRENav). The hydrofoil's vibrations were measured by using two laser doppler velocimetries (LDVs). The LDV measurement points were located at a section that is 95% span length away from the clamped root and at 0.1c or 0.8c from the foil leading-edge on the pressure side of the foil. The measured dry and still-water (zero velocity) natural frequencies were determined by impulse loads generated using an electrodynamic shaker. Blake and Maga's experiments[25]

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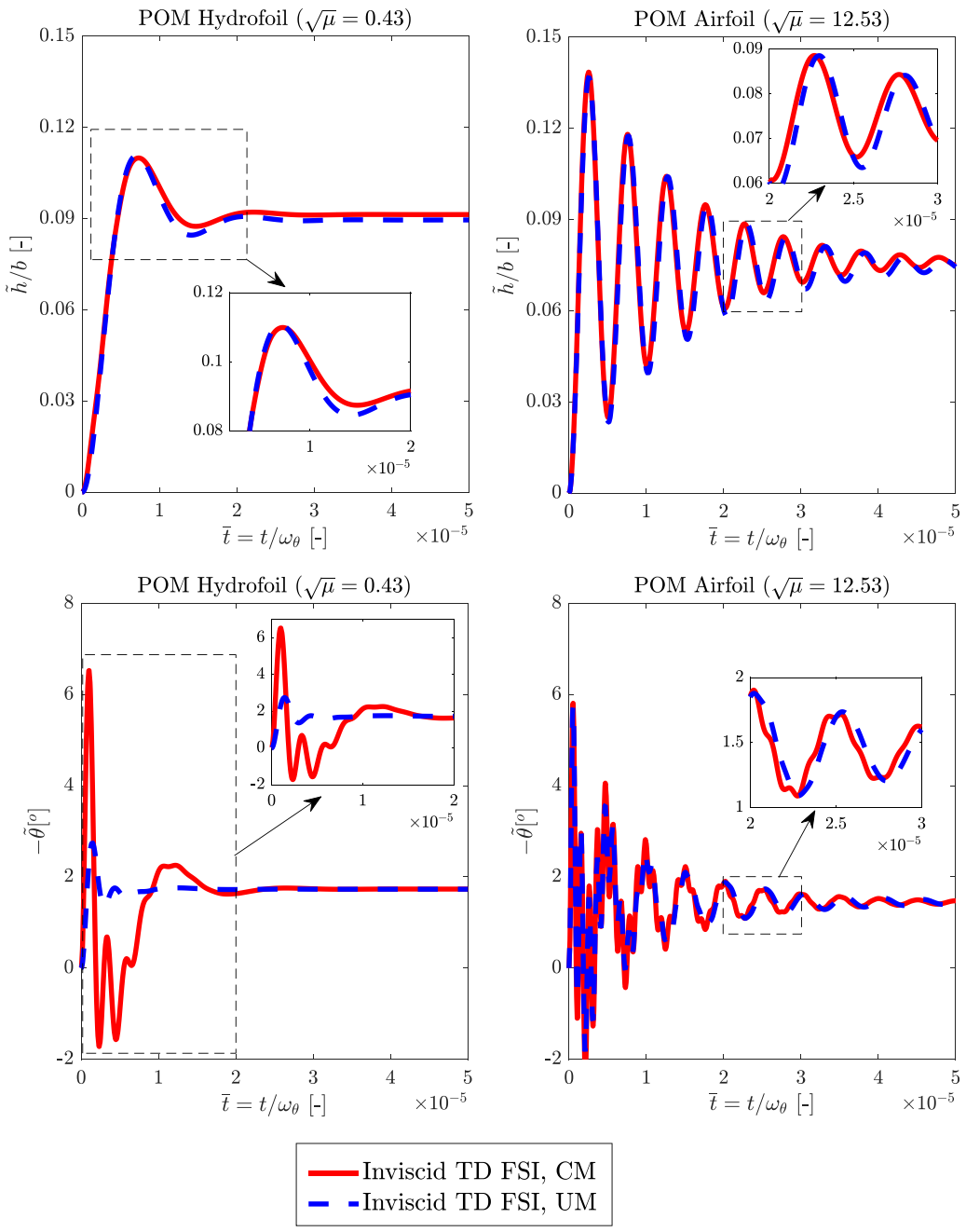


FIG. 5. The time-histories of the bending deformations (top row) and twisting deformations (bottom row) for the inviscid TD FSI model on the cantilevered NACA 0015 POM hydrofoil and POM airfoil at $\bar{U}_{\text{stable}} = 0.2$, AR $= 4$, and $\alpha_o = 8^\circ$.

measured the model responses and loss factors of a rectangular cantilevered hydrofoil (within a bullet cross section shape made of stainless steel, SS). They tested an assortment of hydrofoils with effective aspect ratios of 8, 16, 19, and 27 by changing the span length and chord length. The hydrofoils protruded from a clamp through a 6-in. deadwater region into the open jet of the 12-in. variable-pressure water tunnel at the Naval Ship Research and Development Center. Accelerometer signals were carried by a low-noise coaxial cable, and damping measurements were obtained by filtering decaying acceleration levels using an audio frequency spectrometer. Reese's experiments[29] determined the parameters (e.g., natural frequency and loss factor for a rectangular cantilevered hydrofoil) under the conditions of

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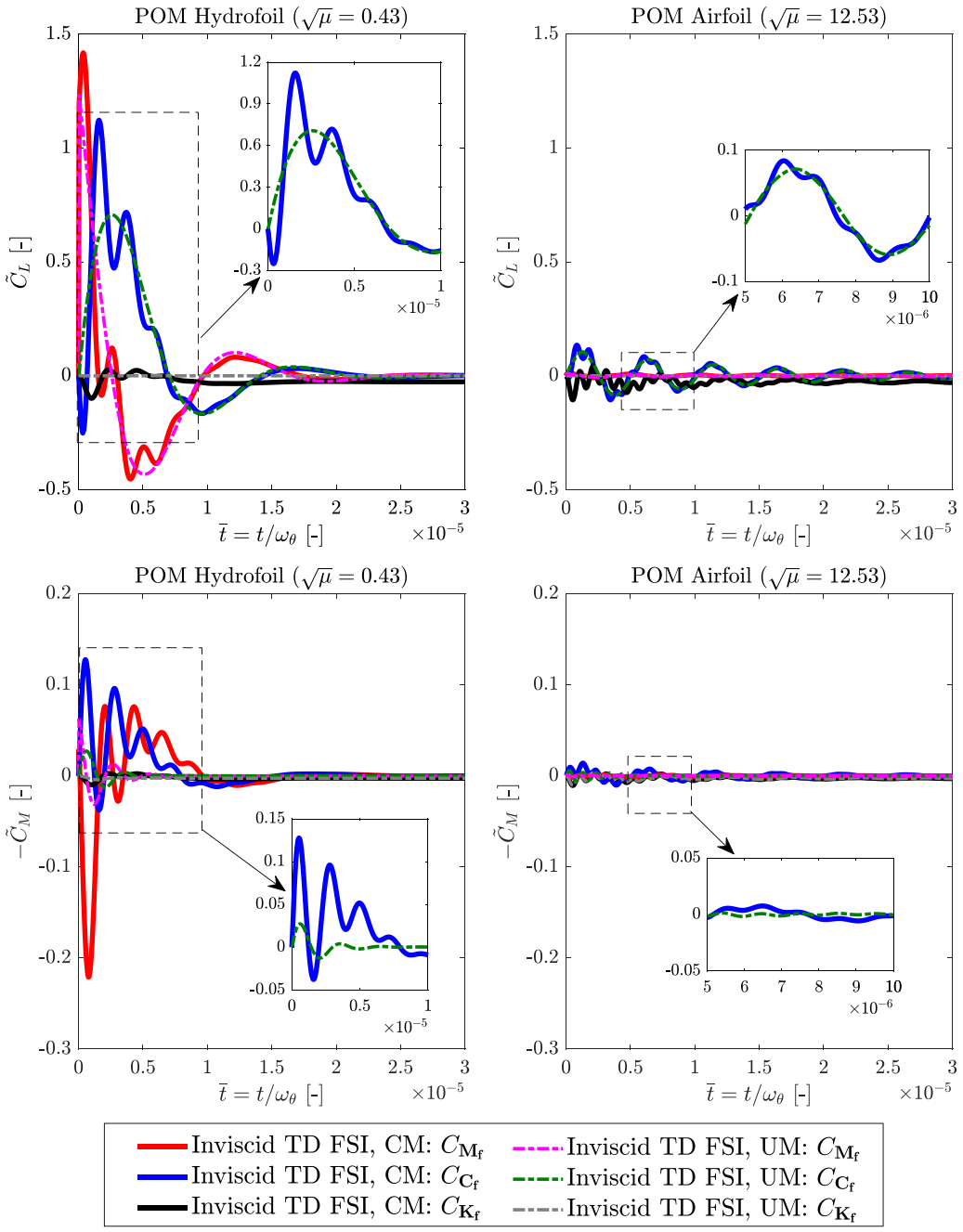


FIG. 6. The time-histories of the fluid inertial, damping, and stiffness components of the lift coefficients (top row) and moment coefficients (bottom row) for the inviscid TD FSI model on the cantilevered NACA 0015 POM hydrofoil and POM airfoil at $U_{\text{stable}} = 0.2$, AR $= 4$, and $\alpha_o = 8^\circ$.

both flow-induced and mechanical forced vibrations with varying angles of attack and free stream velocities. Experimental tests were conducted on the aluminum (Al) NACA 66 hydrofoils with an aspect ratio of 4 at the Penn State University Applied Research Laboratory's 12 in. water tunnel with the rectangular test section. In Reese's experiments, the velocities in experiments were measured by the pressure differential between a static probe and Kiel (total pressure) probe, which was located at equal vertical depth to eliminate the dependence of pressure differential on water column height; the pressures were measured via two Heise DXD 100 psi transducers. The LDV was used as the primary method of capturing vibration velocity measurement in the z-direction. It should be noted that all experimental data are

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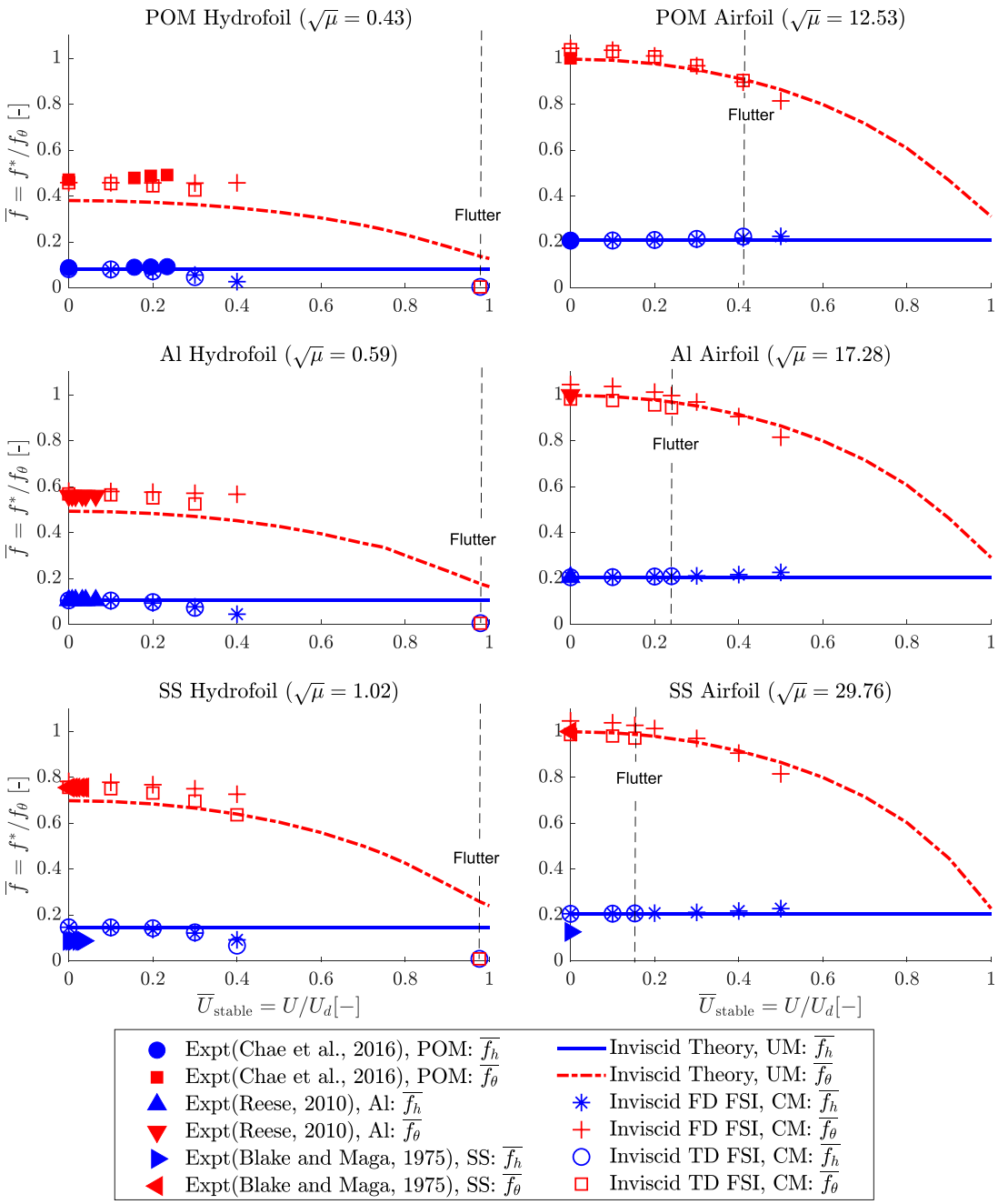


FIG. 7. Non-dimensional bending and twisting natural frequencies ($\overline{f_h} = f_h^*/f_\theta$ and $\overline{f_\theta} = f_\theta^*/f_\theta$) of the cantilevered hydrofoils (left) and airfoils (right) made of different materials [e.g., polyacetate (POM), aluminum (Al), and stainless steel (SS)] at AR = 4 and $\alpha_o = 8^\circ$.

limited to $\bar{U}_{\text{stable}} < 0.25$, which is necessary to avoid material or instability failures in a water tunnel. The Reynolds number ranges of the experiments are 4×10^5–2.1×10^6 for Chae *et al.* experiments,[2] 5.6×10^4–5.6×10^5 for Reese's experiments,[29] and $1 \times 10^5 - 5 \times 10^5$ for Blake and Maga's experiments.[25]

In Figs. 7 and 8, the difference between the inviscid FD and TD FSI solutions is that the inviscid FD FSI solutions assume the simple sinusoidal motion but not explicitly in the inviscid TD FSI solutions. The difference between the inviscid theory results and the inviscid FD/TD FSI results is the flow-induced bend-twist coupling terms, i.e., the off diagonal terms in Eqs. (18)–(20), which are ignored in the inviscid theory results. The differences between the inviscid theory results and experiments are caused by (1) neglection of the flow-induced bend-twist coupling terms in uncoupled mode inviscid simulations and (2)

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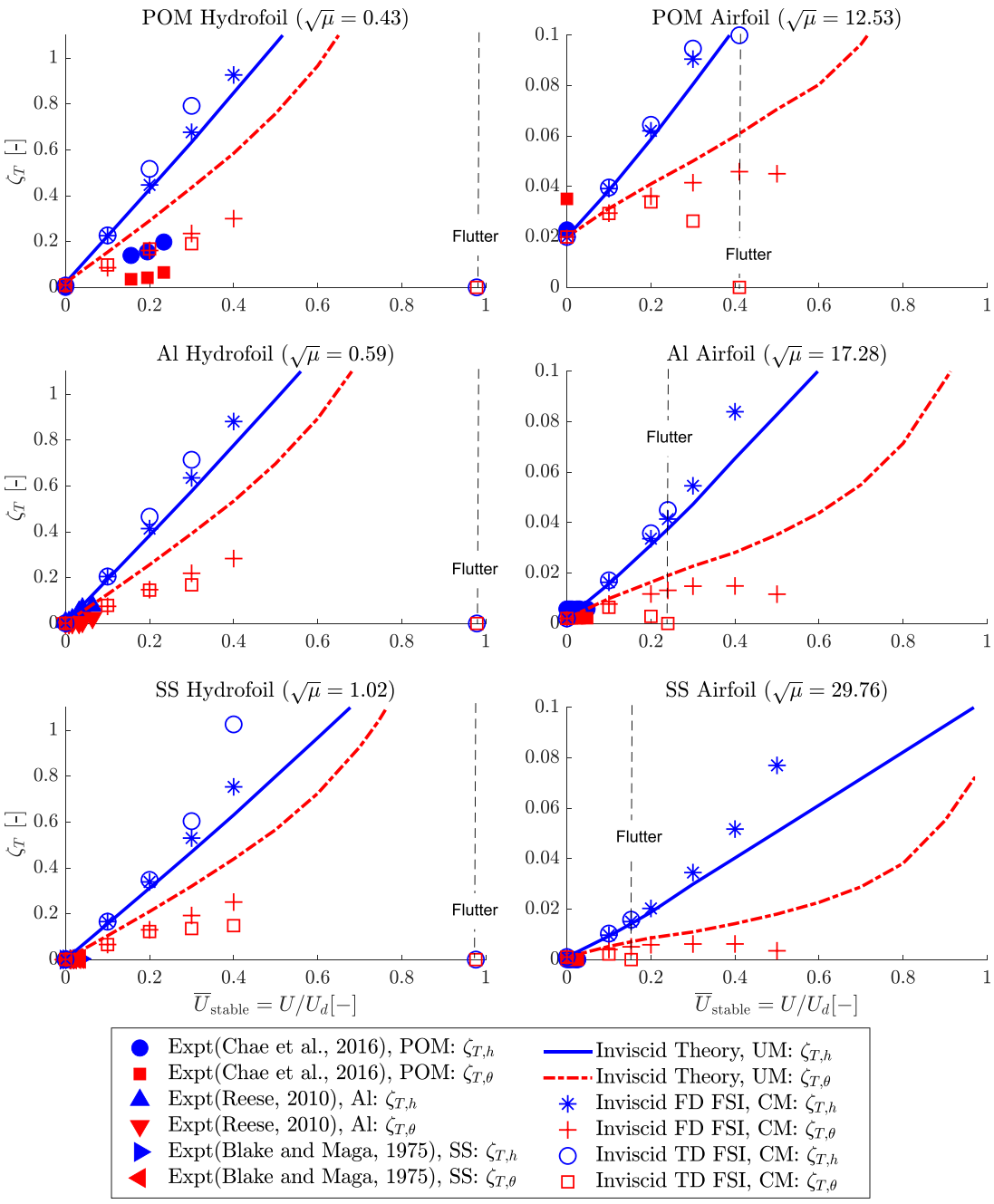


FIG. 8. Total bending and twisting damping coefficients ($\zeta_{T,h}$ and $\zeta_{T,\theta}$) of the cantilevered hydrofoils (left) and airfoils (right) made of different materials [e.g., polyacetate (POM), aluminum (Al), and stainless steel (SS)] at AR = 4 and $\alpha_o = 8^\circ$.

viscous effects. The measured natural frequencies and measured damping values are slightly different from the inviscid predictions, which are expected due to viscous effects. Note that while the frequency of the twisting mode is higher than the bending mode, the damping of the twisting mode is lower than the bending mode. The vertical flutter lines are obtained using the inviscid TD FSI calculations (results shown in the red open square symbols in Fig. 8) when the total damping for the twisting goes to zero, and the total damping is the sum of the solid damping in Table III and fluid damping for a given inviscid analysis.

As shown in Figs. 7 and 8, the inviscid TD FSI simulation provided comparable results in the natural frequency and fluid damping coefficient with much lower central processing unit (CPU)

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requirements. The differences between inviscid and viscous FSI simulations are less than 10% for the natural frequency and fluid damping coefficient shown in the previous work.[3] Therefore, the inviscid FSI simulation provides the computational efficiency necessary to practically explore the general trends of the results and provides a good starting point for the design of any material, geometric, and flow parameters for real-world applications.

Figures 7 and 8 illustrate the dramatic difference in natural frequencies and total damping coefficients between the airfoil and hydrofoil, particularly for the dependence on the inflow velocity. While the natural twisting frequency is reduced with increasing velocity due to increasing the fluid disturbing moment, the natural bending frequency is reduced with velocity due to the flow-induced bend-twist coupling term, $\tilde{\mathbf{K}}_{f_{1,2}}$ in Eq. (20). As the twist deformation is defined as a positive nose down in Fig. 1, the foil undergoes a nose up twist ($\theta < 0$) since the center of pressure (CP) is upstream of the elastic axis (EA), which yields $\tilde{\mathbf{K}}_{f_{1,2}} < 0$, hence the lower effective bending stiffness. The results show that the inviscid theory solution ignores the flow-induced bend-twist coupling effect, and hence, it cannot capture this trend and results in the constant natural bending frequency prediction as the velocity increases. It is worth pointing out that the inviscid theory (*uncoupled mode*, UM) solution cannot predict flutter because it ignores the flow-induced bend-twist coupling effect, which is responsible for driving the twisting damping coefficient to zero. Also, the inviscid theory predictions significantly over-predict the twisting damping, which is dangerous as that implies that actual vibrations and dynamic load amplification may be higher than the prediction. It could lead to accelerated fatigue and flutter that cannot be predicted with the inviscid UM theory simulations. The results also show that the natural frequencies of the hydrofoil are much lower than those of the airfoil due to the fluid added mass, and the in-water to in-air frequency ratio decreases with lighter materials, i.e., lower μ, as suggested in Eqs. (27) and (28). However, the fluid damping coefficients in water are much higher than those in air due to the high fluid density. Also, the fluid disturbing force term is proportional to the fluid density and velocity square. For hydrofoils, the fluid disturbing force grows much quicker with velocity, which allows the mean deformations to increase with velocity while the flow-induced vibrations tend to be rapidly damped out because of the high hydrodynamic damping. Therefore, as the velocity increases, the hydrofoils become increasingly susceptible to divergence, as flutter is typically not an issue because the damping coefficient tends to grow with the velocity [unless the negative off diagonal term in the damping matrix in Eq. (19) becomes significant]. However, for airfoils, flutter tends to occur before the divergence.

D. Influence of the aero/hydro-elastic deformations

The non-dimensional generalized aero/hydro-elastic bending and twisting deformations ($\tilde{h}_{\text{elastic}}/b$ and $\tilde{\theta}_{\text{elastic}}$) shown in Fig. 9 are calculated using Eqs. (37) and (38) with the reduced stable velocity ($\bar{U}_{\text{stable}} = U/U_d$) for cantilevered NACA 0015 foils at $\alpha_o = 8°$ by varying the aspect ratio (AR) and reduced frequency (k). The non-dimensionalized results in Fig. 9 are the same regardless of the operating fluid properties (e.g., water or air) and material properties (e.g., POM, CFRP, aluminum, or stainless steel).

In general, the non-dimensional generalized aero/hydro-elastic deformations increase as the velocity increases. Since the divergence is governed in the hydrofoil, the reduced stable velocity, $\bar{U}_{\text{stable}}$, must be less than one to avoid the divergence, as shown in Fig. 7 (and needs to be even lower including any factor of safety); this will, in turn, limit the maximum bending and twisting deformations. For the airfoil, flutter is the governing instability mechanism, and the maximum deformation will be similarly limited by the corresponding limiting speeds. As the aspect ratio (AR) or reduced frequency (k) increases, the aero/hydro-elastic bending and twisting deformations are increased for a given velocity. At the low velocity (i.e., $\bar{U}_{\text{stable}} \leq 0.2$), the reduced frequency and aspect ratio effects can be ignored for inviscid flows, but this may not be true for viscous flows. It should be noted that the anisotropic composites[10,30] can be used to achieve the higher twist deformations through the material bend-twist coupling. Also, since the bending deformation does not change the elastic aero/hydrodynamic loads, the twist deformation is the critical hydrodynamic performance driver.

IV. CONCLUSIONS AND FUTURE WORK

In this work, parametric studies on the steady and dynamic responses of the spanwise flexible foils were conducted to derive the general critical design and performance parameters of airfoils vs hydrofoils. Rectangular cantilevered foils were considered as a canonical representation of a typical flexible lifting surface with spanwise bending and twisting flexibilities. The air/hydrofoils made of different materials were derived assuming isotropic, linear elastic material responses in potential flow and deeply submerged cases without the Froude number and cavitation number effects. The solid response of the NACA 0015 foil was simplified as a two-degrees of freedom (2-DOF) model characterized by the tip bending and twisting deformations derived by utilizing spanwise shape functions for the three-dimensional (3D) cantilevered structure. For the sake of simplicity and efficiency in order to explore the sizable parametric space, the inviscid theory model ignoring flow-induced bend-twist coupling terms and inviscid frequency- (FD) and time-domain (TD) fluid-structure interaction (FSI) model that accounts for flow-induced bend-twist coupling terms were used to determine the influence of governing non-dimensional material, geometric, and flow parameters.

All components of fluid forces were proportional to the fluid density, and fluid damping forces grew with the velocity while fluid disturbing forces grew with velocity square. Flexible hydrofoils had much lower natural frequencies and higher damping coefficients than flexible airfoils due to higher fluid inertial and damping forces, both of which were proportional to the fluid density. The flow-induced bend-twist coupling effects grew with increasing fluid density and flow speed, and hence such effects were significant on the natural frequencies and damping coefficients of flexible hydrofoils, particularly for the twisting mode, while these effects tend to be negligible for airfoils. Ignoring the flow-induced bend-twist coupling terms leads to over-prediction on the damping, which is dangerous as that implies that actual vibrations and dynamic load amplification may be higher than the prediction and could lead to accelerated fatigue. Moreover, uncoupled mode calculations cannot predict flutter, as the flow-induced bend-twist is responsible for the reduction in fluid damping with higher speeds. The fluid damping coefficients in water were much higher than those in air; hence, the divergence tended to be the governing instability mode for flexible hydrofoils, while flutter tends to be a governing instability mode for flexible airfoils. The maximum aero/hydro-elastic bending and twisting deformations were limited by

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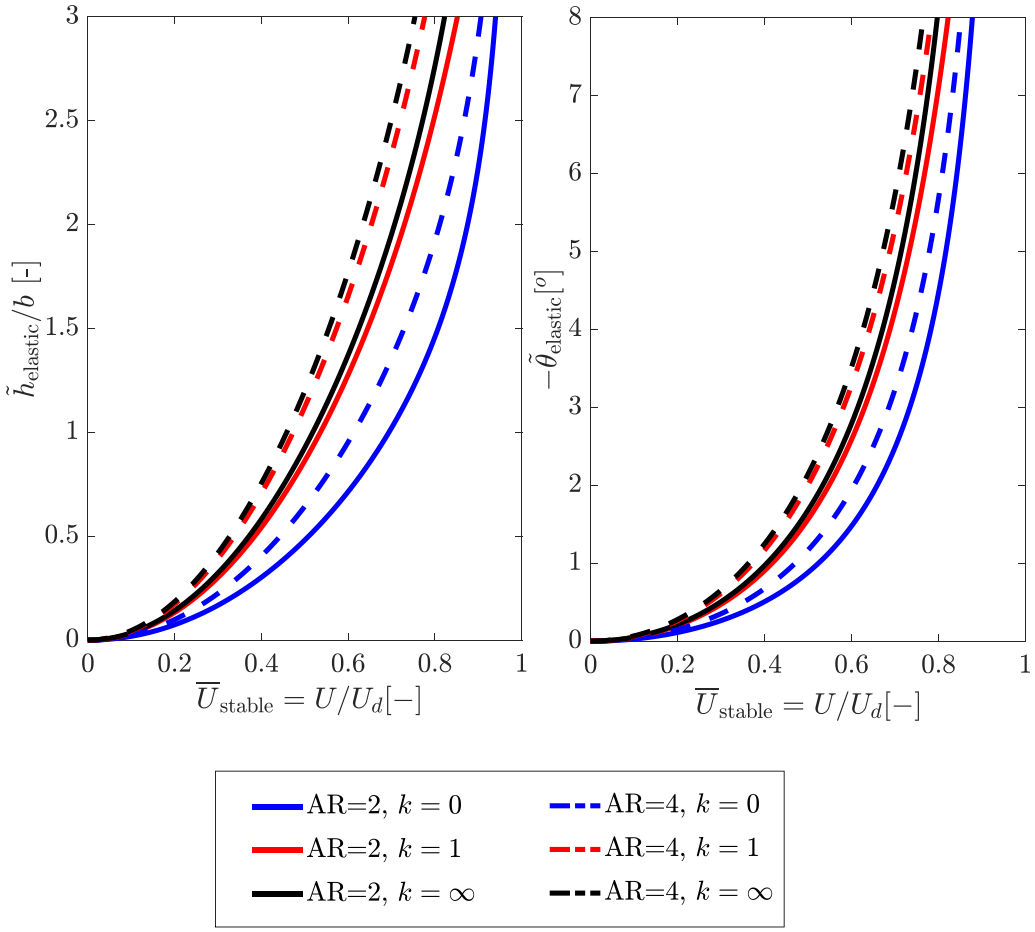


FIG. 9. The non-dimensional generalized aero/hydro-elastic bending deformations ($\tilde{h}_{\text{elastic}}/b$, left) and twisting deformations ($\tilde{\theta}_{\text{elastic}}$, right) calculated using Eqs. (37) and (38) with the reduced stable velocity ($\overline{U}_{\text{stable}} = U/U_d$) of cantilevered NACA 0015 foils at $\alpha_o = 8^\circ$ for the different AR $= 2, 4$ and $k = 0, 1, \infty$.

the corresponding reduced stable velocity to avoid the divergence or flutter.

For future work, viscous FSI simulations, including nonlinear and inelastic material behaviors, are necessary to investigate 3D dynamic response and stability of the flexible foils with large deformations.

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DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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